IRP Research Plan_2026-05-07

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Date: 7 May 2026
Purpose: Supervisor discussion · Paper revision basis

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1. Core Research Question

> How can environmental and contextual design help people read their own body signals more accurately — and provide non-judgmental external reference when those signals are unclear?

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2. Target User

Specific profile:
Adults who want to lose weight or eat more healthily but consistently struggle — they don't know what to eat, resort to restriction (starving themselves), and feel uncertain after eating ("was that okay?"). They need reassurance that they're on the right path. Includes both home cooks and people who primarily eat takeout.

Core insight:
This person needs external validation not because they lack nutritional information, but because they have lost trust in their own body's signals. The problem is **broken interoceptive trust**, not a knowledge gap. The intervention is not more data — it is rebuilding the conditions under which body signals become legible.

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3. Theoretical Framework

Fogg B=MAP — Focused on A + M

Component	Design Direction	Practical Meaning
A (Ability)	Environmental design — reduce friction for healthy choices	Change the physical context so that good choices become the default
M (Motivation)	Non-punitive feedback — build positive felt sense	When body signals are unclear, offer a gentle external reference — not a score, but a prompt to notice

Key position: Body signals can be misleading

- Satiety signals take approximately 20 minutes to travel from gut to brain
- Ultra-processed foods are engineered to bypass satiety mechanisms
- Chronic dieters have disrupted interoceptive signals — hunger and emotional eating become indistinguishable
- Blood sugar fluctuations create false hunger cues unrelated to actual nutritional need

Therefore: the goal is **not** "trust your body completely." It is: *design conditions where body signals become clearer, and provide gentle support when they remain ambiguous.*

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4. Secondary Research Map

Existing Literature (keep in paper)

- Fogg (2009) Behaviour Model (B=MAP)
- Kahneman (2011) Present bias
- Tribole & Resch (2020) Intuitive Eating
- Wallace et al. (2025) Reification of data
- Roth et al. (2024) Self-monitoring and eating disorder risk
- Zeevi et al. (2015) Bioindividuality in glycaemic response
- Thaler & Sunstein (2008) Nudge

Literature to Add (new direction)

Research Area	Key Citation	Core Finding
Eating order	Sun et al. (2020) PATTERN study	Eating vegetables/protein before carbohydrates significantly reduces postprandial glucose peaks
Plate & vessel perception	Van Ittersum & Wansink (2012)	Smaller plates exploit the Delboeuf illusion — same portion looks larger, people naturally serve and eat less
Food visibility & accessibility	Painter, Wansink & Hieggelke (2002)	Placing healthy food in the most visible and accessible position reliably shifts choices
Eating pace & satiety	Robinson et al. (2014) meta-analysis	Slower eating allows gut-brain satiety hormones (GLP-1, CCK) to signal fullness, reducing overall intake
Food arrangement & separation	Szocs & Lefebvre (2017)	Spreading food horizontally rather than stacking increases perceived volume — brain anticipates greater satiety
Micro-environmental	Marteau & Hollands (2015)	Altering food proximity and portion vessel size is a low-cost, high-effect behavioural

Research Area	Key Citation	Core Finding
design	Cochrane review	intervention
Cultural food environments	Rozin et al. (2003)	French people eat less not through willpower but through structurally smaller portions and physical environment
Positive computing	Calvo & Peters (2014)	Framework for designing technology that supports wellbeing and autonomy rather than surveillance

Search databases: Google Scholar, PubMed
Keywords: `food environment design`, `eating order glycemic response`, `portion perception vessel`, `eating rate satiety meta-analysis`, `food arrangement visual perception`, `choice architecture food behaviour`

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5. Primary Research Plan

Participant Criteria

Number: 6–8 adults
Age: 22–45
Inclusion: Experience of trying to change eating habits (regardless of success); prior use of any dietary tracking tool (app, notebook, meal plan)
Exclusion: Current clinical eating disorder diagnosis
Diversity: At least 2 participants who primarily eat takeout; at least 2 from non-Western food cultures where possible

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Phase 1: Kitchen Walkthrough + Semi-Structured Interview

Duration: 60–75 minutes
In person or video call

First 15 minutes: Kitchen / eating space tour

Ask the participant to walk you through their kitchen or main eating space. For remote participants: video call with camera on.

"Open your fridge for me — how is everything arranged?"
"Where do you keep snacks? Where are the vegetables and fruit?"
"What's the first thing you see when you open the fridge?"
"Where do you usually grab food from when you're not really thinking about it?"

> "What food in your kitchen is already prepared and ready to eat? What requires extra effort?"

Rationale: Environmental design research must examine real environments. People's descriptions often differ from reality. A walkthrough reveals what is actually hidden, what is visible, and whether the space structurally supports or undermines healthy choices.

Following 45 minutes: Semi-structured interview

Warm-up (5 min)

> "Walk me through a typical eating day for you — how much is planned in advance versus spontaneous?"
> "Where do you usually eat? What's around you when you eat?"

Finding failure moments (15 min)

> "Think of a time you seriously tried to change how you eat but couldn't sustain it. What specifically made it difficult — too much effort? Felt pointless? Something else?"
> "When do you eat on autopilot, without really thinking? What tends to trigger that — a specific place, an emotion, a time of day?"
> "If you could change one physical thing in your kitchen or eating space — not using willpower, just changing the environment — to make eating well easier, what would it be?"

Exploring body signals (15 min)

> "When you finish a meal that feels 'right', how do you know? What does your body actually feel like ten minutes later?"
> "Have you ever finished a meal feeling uncertain — not sure whether it was okay? What does that uncertainty feel like?"
> "Can you remember a moment when you clearly felt 'that's enough' or 'I ate well'? What caused that feeling?"

Tracking tool experience (10 min)

> "Have you ever used an app or any tool to track what you eat? What did those numbers or records actually do for you?"
> "When did tracking feel like support, and when did it feel like judgement or surveillance?"
> "If something could tell you a little about how your meal went — without using numbers — what kind of feeling or information would you want?"

Phase 2: 3-Day Post-Meal Photo Diary

Timing: Conducted after Phase 1, over 3 consecutive days
Frequency: After each main meal

Instructions for participants:

After each main meal, complete the following within 60 seconds:

① **Take one photo**
Not of the food — of the **aftermath**: the empty plate, the table, the surroundings, your hands, the view from where you're sitting.

② **Write three lines** (phone notes or paper, either is fine)

> *Where am I eating? What else am I doing right now?*
> *What does my body feel like right now?* (Use descriptive words, no numbers: heavy, light, warm, bloated, calm, anxious, satisfied, hollow, settled...)
> *Did this meal feel okay? Why, or why not?*

Rationale:
Captures real eating moments, not retrospective memories. The "aftermath" photo — rather than a food photo — reveals the actual eating environment and context. Photos become the opening material for Phase 3.

Phase 3: Design Probe + Reaction Session

Duration: 30–45 minutes
Conducted after the diary period

Opening: Use the participant's own diary photos to begin the conversation — *"Looking at this photo — what were you thinking at that moment?"*

Then present two design concepts (simple sketches or written descriptions):

Concept A — Eating Structure Intervention (Environmental Design)

Description:
At mealtimes, food is arranged in separate sections on the plate — vegetables in one area, protein in another, carbohydrates in a third. No mixing or stacking. Eating order follows: vegetables first, then protein, then carbohydrates. No phone on the table during the meal.

Probe questions:

> "Is this realistic in your daily life?"
> "Which part would you genuinely try? Which part would feel frustrating or unnecessary?"

"If you ate like this, do you think the experience of finishing the meal would feel different?"
"What would stop you from doing this?"

Concept B — Post-Meal Body Check-In (Non-numeric Feedback)

Description:
Twenty minutes after finishing a meal, a simple cue — a phone vibration, a small ambient light, a brief prompt — invites you to pause and notice how your body feels right now. No score. No judgement. Just: *"What does your body feel like at this moment?"*

Probe questions:

"Would this feel useful, or would it feel intrusive?"
"If this existed, what form would you want it to take — a phone notification? A physical object? Something else?"
"Would it feel like surveillance, or like care?"
"Would you actually stop and notice, or would you dismiss it?"

Expected Data Outputs

Method	What you get
Kitchen walkthrough	Real environment data: what is visible, what is hidden, whether the space structurally supports healthy choices
Interview	First-person accounts of failure moments, emotional states, body signal vocabulary in participants' own words
3-day diary	Real-time capture across three days — not retrospective; visual record of actual eating contexts
Design probe	Acceptance and resistance to two intervention directions; preferred vocabulary and forms of feedback

Analysis Method

Reflexive thematic analysis (Braun & Clarke, 2006).
Coding dimensions:

At which moment failure occurs
Emotional state and body signal vocabulary
Acceptance of and resistance to environmental interventions

Preferences and fears around feedback forms

Synthesis: translate participant language directly into design principle vocabulary. Avoid re-quantifying qualitative findings.

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6. Expected Contribution

A set of **environment and feedback design principles** for dietary behaviour, answering:

- At which specific moments is intervention most effective and lowest friction?
- Which environmental changes have low effort but real behavioural effect?
- What forms of feedback feel supportive rather than judgmental or surveillance-like?